

## 7 Scattering theory

Quantum scattering theory plays an essential role in describing chemical reactions and photoionization. Although all of these phenomena are time dependent, scattering theory is most accessible from a time-independent perspective. Here we keep things simple by treating one-dimensional problems.

### 7.1 Scattering states

In a previous lecture we studied the free-particle states  $|p\rangle$  which are eigenfunctions of the free-particle Hamiltonian. A more general Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}), \quad (7.1)$$

where  $V(x) \rightarrow 0$  in the limit  $x \rightarrow \pm\infty$ .<sup>\*</sup> Such a barrier is represented in Figure 6. A particle incoming from the left with momentum  $p$  will have energy  $E = p^2/2m$  and will either be reflected or transmitted over/through the barrier. At large distances from the barrier, the potential is zero and thus we know that the wave function will take the form of the free-particle states discussed above.

The scattering states are time-independent and thus one must consider the incoming, reflected and transmitted parts all together as well as the more complex wave function which exists in the barrier region. These states solve

$$\hat{H} |\psi_p\rangle = E |\psi_p\rangle, \quad (7.2)$$

where the energy is still  $E = p^2/2m$ , where  $p$  refers to the initial momentum. Although the energy  $E$  is a conserved quantity, the momentum operator no longer commutes with the Hamiltonian and thus we can only specify the momentum at the asymptotes where the potential is flat.

<sup>\*</sup>Actually all this theory can easily be generalized to treat exo- or endothermic reactions too. The potential simply needs to become flat at the asymptotes.

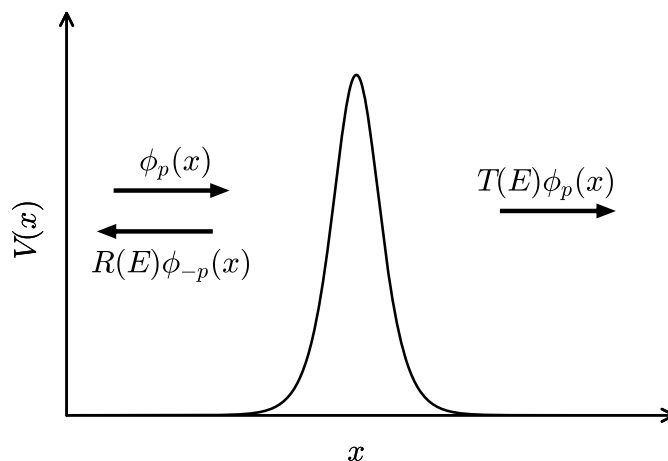


Figure 6: Schematic for scattering state for particle incoming from the left

At these asymptotes we know the values of the momentum and can thus write

$$\lim_{x \rightarrow -\infty} \psi_p(x) = \phi_p(x) + R(E)\phi_{-p}(x) \quad (7.3)$$

$$\lim_{x \rightarrow \infty} \psi_p(x) = T(E)\phi_p(x). \quad (7.4)$$

To find the values of the complex numbers  $R(E)$  and  $T(E)$ , we must solve the time-independent Schrödinger equation which must usually be performed numerically. However, for a couple of cases it can be solved analytically such as for the square barrier, which you have probably solved in previous courses.\*

## 7.2 Flux and rate

The probability of the particle reflecting is given by  $|R(E)|^2$  and the probability of it being transmitted is  $|T(E)|^2$ . Of course, probabilities must sum to 1 so  $|R(E)|^2 + |T(E)|^2 = 1$ .

Flux through point  $x = s$  is Heisenberg time derivative of Heaviside step function,  $\theta(x - s)$ ,

$$\hat{F}(s) = \frac{i}{\hbar} [\hat{H}, \theta(\hat{x} - s)]. \quad (7.5)$$

Exercise: using this definition, show that

$$\hat{F}(s) = \frac{1}{2m} [\delta(\hat{x} - s)\hat{p} + \hat{p}\delta(\hat{x} - s)]. \quad (7.6)$$

We can calculate the flux through a point,  $s$ , far to the right of the barrier,

$$\langle \psi_p | \hat{F}(s) | \psi_p \rangle = |T(E)|^2 \langle p | \hat{F}(s) | p \rangle \quad (7.7)$$

but the flux is everywhere the same and thus also holds if  $s$  is at the barrier or even on its left.

The flux of a free particle through a point,  $x = s$ , is

$$\langle p | \hat{F}(s) | p \rangle = \frac{1}{2m} [\langle p | \delta(\hat{x} - s) \hat{p} | p \rangle + \langle p | \hat{p} \delta(\hat{x} - s) | p \rangle] \quad (7.8a)$$

$$= \frac{1}{2m} \int_{-\infty}^{\infty} [\langle p | x \rangle \langle x | \delta(\hat{x} - s) \hat{p} | p \rangle + \langle p | \hat{p} \delta(\hat{x} - s) | x \rangle \langle x | p \rangle] dx \quad (7.8b)$$

$$= \frac{1}{2m} \int_{-\infty}^{\infty} [\langle p | x \rangle \delta(x - s) \langle x | p \rangle p + p \langle p | x \rangle \delta(x - s) \langle x | p \rangle] dx \quad (7.8c)$$

$$= \frac{1}{2m} [\langle p | s \rangle \langle s | p \rangle p + p \langle p | s \rangle \langle s | p \rangle] \quad (7.8d)$$

$$= \frac{p}{m} (2\pi\hbar)^{-1} e^{+ips/\hbar} e^{-ips/\hbar} \quad (7.8e)$$

$$= \frac{1}{2\pi\hbar} \frac{p}{m} \quad (7.8f)$$

The probability of reaction is given by the ratio of transmitted flux to incoming flux

$$P(E) = \frac{\langle \psi_p | \hat{F}(s) | \psi_p \rangle}{\langle p | \hat{F}(s) | p \rangle} \quad (7.9)$$

$$= |T(E)|^2. \quad (7.10)$$

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\*If not, look at wikipedia!

Therefore, to obtain  $P(E)$ , we just need to know the flux of the wavefunction,  $|\psi_p\rangle$  at a point,  $x = s$ .

The thermal rate constant for the reaction is found by a Boltzmann average over the probability of reaction,

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_0^\infty P(E) e^{-\beta E} dE, \quad (7.11)$$

where for a 1D system, the reactant partition per unit length is

$$Q_r(T) = \sqrt{\frac{m}{2\pi\beta\hbar^2}}. \quad (7.12)$$

### 7.3 Tunnelling effect

In classical mechanics  $P(E) = \theta(E - V^\ddagger)$  as only particles above the barrier can be transmitted and they all pass over. However, due to tunnelling effects, this is not true in quantum mechanics. The thermal rate constant for a classical particle in one-dimension is given by

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_{V^\ddagger}^\infty e^{-\beta E} dE \quad (7.13)$$

$$= \frac{1}{2\pi\beta\hbar Q_r(T)} e^{-\beta V^\ddagger}. \quad (7.14)$$

We have therefore derived the one-dimensional version of the TST formula, equation (1.33), presented in the first lecture.

By horrendous mathematical analysis (which these days would be done by Mathematica) Bell was able to obtain the analytical expression for  $P(E)$  for a parabolic barrier  $V(x) = V^\ddagger - m\omega^2 x^2/2$  which is

$$P(E) = [1 + \exp(2\pi(V^\ddagger - E)/\hbar\omega)]^{-1} \quad (7.15)$$

From this we note that at an energy close to the barrier top, only half of the flux is transmitted.

For a parabolic barrier, the rate is

$$k(T) = \frac{1}{2\pi\hbar Q_r(T)} \int_{-\infty}^\infty P(E) e^{-\beta E} dE \quad (7.16)$$

$$= \frac{1}{2\pi\beta\hbar Q_r(T)} \frac{\frac{1}{2}\beta\hbar\omega}{\sin \frac{1}{2}\beta\hbar\omega} e^{-\beta V^\ddagger} \quad (7.17)$$

where we have used Mathematica to perform the integral.

This gives us a simple method for estimating the shallow tunnelling effects on any barrier which has an approximately parabolic shape near its top. However, note that the expression blows up when  $\beta > 2\pi/\hbar\omega$  and thus cannot describe deep tunnelling occurring at low temperatures where the real barrier will deviate from the parabolic approximation.

A better approximation can be obtained from the Eckart barrier,

$$V(x) = \frac{V^\ddagger}{\cosh^2(x/a)}$$

for which a semiclassical approximation to the reaction probability is

$$P(E) \sim \frac{1}{1 + \exp W(E)/\hbar} \quad (7.18a)$$

$$\sim e^{-W(E)/\hbar} \quad (7.18b)$$

where  $W(E) = 2\hbar\alpha(1 - \sqrt{E/V^\ddagger})$  and  $\alpha = \pi\sqrt{2ma^2V^\ddagger}/\hbar$  and the second approximation is valid only for  $E \ll V^\ddagger$ . Note that Eckart gives the complicated exact expression in his 1930 paper.

An analytic formula for the thermal rate of the Eckart barrier in the deep-tunnelling regime can be obtained using another approximation: the steepest-descent integration.

$$k(T) \sim \frac{1}{2\pi\hbar Q_r(T)} \int e^{-W(E)/\hbar - \beta E} dE. \quad (7.19)$$

Expanding the exponent as a Taylor series around its minimum  $E = E^*$ , defined such that  $W'(E^*) = -\beta\hbar$ , gives

$$k(T) \sim \frac{1}{2\pi\hbar Q_r(T)} \int e^{-W(E^*)/\hbar - W'(E^*)(E-E^*)/\hbar - W''(E^*)(E-E^*)^2/2\hbar - \beta E} dE \quad (7.20a)$$

$$= \frac{1}{2\pi\hbar Q_r(T)} e^{-S/\hbar} \int e^{-W''(E^*)(E-E^*)^2/2\hbar} dE \quad (7.20b)$$

$$= \frac{1}{2\pi\hbar Q_r(T)} \sqrt{\frac{2\pi\hbar}{W''(E^*)}} e^{-S/\hbar}. \quad (7.20c)$$

where  $S = W(E^*) + \hbar\beta E^*$ .

From this you can derive

$$E^* = \frac{\alpha^2}{\beta^2 V^\ddagger} = \frac{2\pi^2 m a^2}{\beta^2 \hbar^2},$$

which means that reactions will tunnel at low energies when the temperature is low and the barrier is thin.

Exercise: make Arrhenius plots for the Eckart barrier. Note that these expressions are only valid below a certain temperature,  $\beta > \alpha/V^\ddagger$ , when the tunnelling mechanism dominates, i.e. when  $E^* < V^\ddagger$ . Using the more accurate expression, equation (7.18a), you can obtain the rate by numerical integration. This is valid at all temperatures and should be in agreement with the classical rate at high temperatures.

## 7.4 Further Reading

The following textbooks will give you further information on the topics of this lecture:

*More on momentum states*

- Feynman Lectures Volume III chapter 16, <https://www.feynmanlectures.caltech.edu>

*Hard-core theory of scattering (for future reference only)*

- Mark Child, Molecular collision theory

- John Taylor, Scattering theory: The Quantum Theory of Nonrelativistic Collisions (1972)

*To learn more about tunnelling effects in chemical reactions, see my articles:*

- “Understanding Chemical Reactions beyond Transition-state Theory”, Chimia **72**, 309-312 (2018)
- “Perspective: Ring-polymer Instanton Theory”, J. Chem. Phys. **148**, 200901 (2018)